# Healthcare Provider Utilization & Payment Analysis

Medicare Provider Utilization, Cost & Payment Pattern Analysis

*Python/Pandas • SQL • Power BI*

## Business Question

Healthcare management wants to understand provider utilization and payment patterns to identify insignificant values and unusual patterns that may require further investigation.

## Overview

This project analyzes a Medicare Physician Provider and Service public dataset to identify which contains provider-level information such as provider identifiers, provider type, location, and demographic details, along with utilization and financial measures such as total beneficiaries, total services, beneficiary-day services, submitted charges, and Medicare payment amounts.

The analysis was conducted using Python/Pandas for data cleaning and validation, SQL for provider-level, provider type-level, and peer-benchmarking analysis, and Power BI for interactive dashboard development and communication of findings to non-technical stakeholders.

## Dataset

- Source: Medicare Physician Provider and Service public dataset
- 292,663 provider-service records across 29 columns
- Key fields: Provider NPI, provider type, HCPCS service code, HCPCS description, total services, total beneficiaries, average submitted charge, average Medicare allowed amount, average Medicare payment, average standardized amount

## Tools Used

- Python (pandas, numpy): Data cleaning, missing-value handling, numerical validation
- SQL: Provider-level, provider-type-level, Provider-type level and peer-benchmarking analysis
- Power BI: Final interactive dashboard design

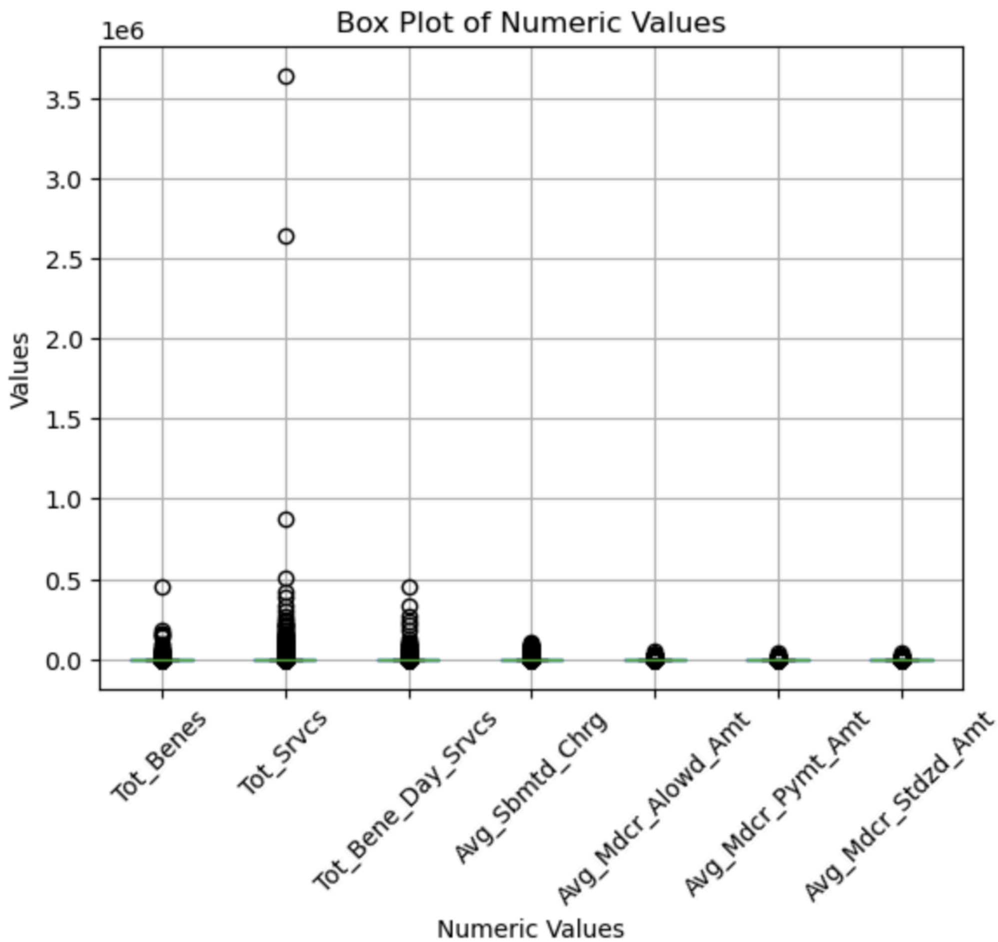
## Methodology

**Python**

Used Python with Pandas to examine the dataset structure, data types, missing values, and distributions of key variables.

1. **First Name:** 5.53% of values were missing. Investigation showed that the missing values were associated with organizational providers, where a first name is not applicable. One individual provider had a missing first name, so it was flagged using first_name_missing_individual while retaining the original NULL value.
2. **Middle Name:** 34.37% of values were missing. Since a middle name is not necessarily applicable to every individual, the missing values were considered acceptable and retained as NULL.
3. **Provider Credentials:** 10.63% of values were missing. Investigation showed that 16,193 missing values belonged to organizational providers, where credentials are not applicable, so they were retained. Another 14,920 missing values belonged to individual providers and were retained but flagged using credential_missing_individual for further investigation.
4. **Gender:** 5.53% of values were missing, representing 16,193 records. All of these records belonged to organizational providers and also had missing first names. Therefore, the missing gender values were considered expected rather than a data-quality issue.
5. **Street Address 2:** Removed this column because 73.86% of its values were missing and it was not relevant to the provider utilization and payment analysis.
6. **RUCA and RUCA Description:** Retained these fields despite a small number of NULL values because the fields were not central to the analysis and the missingness was limited.
7. **ZIP Code:** Identified inconsistent/mixed data types during profiling. The ZIP Code field was ultimately dropped because it was not required for the project's utilization and payment analysis.
8. **Numeric variables:** Examined the distributions of Tot_Benes, Tot_Srvcs, Tot_Bene_Day_Srvcs, Avg_Sbmtd_Chrg, Avg_Mdcr_Alowd_Amt, Avg_Mdcr_Pymt_Amt, and Avg_Mdcr_Stdzd_Amt. All showed strong right-skewed distributions, with most providers having relatively lower values and a smaller number having extremely high values.
9. **Potential outliers:** Used distribution analysis and boxplots to identify extreme observations. These values were retained rather than automatically removed, because high values can represent legitimate high-volume providers or high-cost services. They were treated as potential areas for further investigation during the analysis.

In [37]:
```
df[numeric_column].boxplot(rot=45)
plt.title('Box Plot of Numeric Values')
plt.xlabel('Numeric Values')
plt.ylabel('Values')
plt.show()
```




I didn't simply delete missing values or outliers. I investigated them first and made decisions based on whether they were expected, relevant to the analysis, or required further investigation.

**SQL**

The analysis was performed at both the **provider-type level and provider level**, focusing on service utilization and payment patterns.

1. **Provider-Type Level Analysis**

   **1.1 Provider type and service utilization**

I found that Clinical Laboratory had the highest service utilization, with approximately 13.55 million services, or 11,364 services per provider, which was more than three times the second-highest provider type, Ambulance Service Provider.

More than half of Clinical Laboratory's service volume came from two services: COVID-19 testing (4.27 million services) and P9603 (2.77 million services).

I then examined the payment associated with these high-volume services and found that high service volume did not necessarily result in the highest payment per service.

Estimated Medicare Payment = Avg_Mdcr_Pymt_Amt × Tot_Srvcs

**1.2 Provider type and payment pattern**

I found that provider types have different utilization and payment structures. Some provider types have high service utilization with relatively lower payment per service, while others have lower utilization but substantially higher payment per service.

## 2. Provider-Level Analysis

**1.2 Provider and service utilization**

Vcare Testing Centre Corp. recorded the highest provider-level service volume at approximately 3.64 million services. Its high volume was primarily driven by COVID-19 testing, generating an estimated $42.8 million in Medicare payments.

**1.2 Provider and payment pattern**

Caris MPI, Inc. generated the highest estimated Medicare payment among providers, at approximately $60.2 million, despite having only 20,456 services, which was much lower than some of the highest-volume providers.

## 3. Peer Benchmarking

I benchmarked each provider's payment per service against the average payment per service of other providers of the same provider type. I used a minimum threshold of 100 services so that providers with very few services would not significantly affect the benchmark.

The provider-type benchmark was calculated as: Total estimated Medicare payment ÷ Total services.

The analysis identified Caris MPI, Inc. as a potentially unusual provider. Caris had approximately 20,456 services, an estimated $60.2 million in Medicare payments, and approximately $2,942 payment per service, which was around 165× its provider-type benchmark.

I then drilled down into Caris's services and found that HCPCS 81479 (Molecular Pathology Procedure) was the primary driver of its high estimated payment. I compared Caris with the other provider performing the same HCPCS code and found that Caris's estimated payment per service was approximately $2,942 compared with $705, or about 4.2× higher.

Because only two providers performed HCPCS 81479 in the dataset, I treated this as a potentially unusual payment pattern requiring further investigation, rather than evidence of inappropriate payment or fraud.

## Appendix — Key Results

| Analysis | Metric | Result |
|---|---|---|
| Provider Type → Utilization | Clinical Laboratory total services | 13.55M |
| Clinical Laboratory → Service Drill-down | COVID-19 testing (K1034) services | 4.27M |
| Provider Type → Payment | Clinical Laboratory payment/service | $17.84 |
| Provider → Utilization | Vcare Testing Centre Corp. total services | 3.64M |
| Provider → Payment | Caris MPI estimated Medicare payment | $60.2M |
| Peer benchmarking | Caris MPI payment/service | $2,942 |
| Peer Benchmarking | Clinical Laboratory benchmark | $17.80 |
| Peer Benchmarking | Caris MPI peer ratio | 165× |
| Same-Service Comparison | Gravity Diagnostics payment/service (81479) | $705 |
| Same-Service Comparison | Caris MPI payment/service (81479) | $2,942 |

## Power Bi

I used **Power BI** to present the key utilization and payment findings interactively and allow users to compare provider types and investigate potentially unusual provider-level patterns.

- I used Power Query Editor to create an analysis table from the SQL query results for further analysis in Power BI.
- I created DAX measures, KPI cards, charts, and tables to visualize key utilization and payment metrics and highlight the findings identified during the SQL analysis.

## Business Recommendation

1. Further investigate providers with unusually high payment per service compared with their peers.
2. Review high-volume providers and services to understand what is driving their utilization.
3. Use peer benchmarking as a way to identify providers that may need additional review.
4. For cases like Caris MPI, use claim-level and reimbursement data to understand why the payment differs from other providers performing the same service.

## Conclusion

The analysis showed that service utilization and payment were quite different across provider types and providers. Clinical Laboratory had the highest service volume, mainly driven by COVID-19 testing and P9603. However, having a high number of services did not always mean having a high payment per service.

At the provider level, Vcare Testing Centre Corp. had the highest service volume, while Caris MPI had the highest estimated Medicare payment despite having much fewer services. When I compared Caris with other Clinical Laboratory providers, its payment per service was much higher than the peer benchmark. I then found that HCPCS 81479 was mainly driving this difference, and Caris also had a higher payment per service than the other provider performing the same service.

Overall, the analysis helped identify **unusual utilization and payment patterns that could be investigated further**. These results do not mean that the payments were inappropriate or fraudulent.

## Dashboard

This interactive dashboard provides an overview of providers, services, utilization, and medicare payment, along with provider-type comparisons and unusual payment patterns.